ELITE IGCSE ACADEMY

Student Past Paper Solutions

Jun 2023 P2H Worked Solutions

Jun 2023 | P2H

31 questions ▪ question image followed by worked solution

PREPARED BY

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STUDENT EDITION

Worked answers included

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PAST PAPER QUESTION

Q#: 1 Marks: 3

TOPIC: **Algebraic Proof**

Jun 2023 P2H - Jun 2023 | P2H

1 Show that $4\frac{2}{3} \div 1\frac{1}{5} = 3\frac{8}{9}$

(Total for Question 1 is 3 marks)

PAST PAPER SOLUTION

Q#: 1 Marks: 3

TOPIC: **Algebraic Proof**

Jun 2023 P2H - Jun 2023 | P2H

WORKED SOLUTION

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1. Evaluate fraction

$$4\frac{2}{3} = \frac{14}{3}, \qquad 1\frac{1}{5} = \frac{6}{5}$$

$$4\frac{2}{3} \div 1\frac{1}{5} = \frac{14}{3} \div \frac{6}{5}$$

$$= \frac{14}{3} \times \frac{5}{6} = \frac{70}{18} = \frac{35}{9}$$

$$\frac{35}{9} = 3\frac{8}{9}$$

FINAL ANSWER

$3\frac{8}{9}$.

PAST PAPER QUESTION

Q#: 2 Marks: 3

TOPIC: **Probability Toolkit**

Jun 2023 P2H - Jun 2023 | P2H

2 A biased spinner can land on green or on yellow or on brown or on pink.

The table gives the probabilities that, when the spinner is spun, it will land on green or on yellow or on brown.

Colour	**green**	**yellow**	**brown**	**pink**
Probability	0.32	0.13	0.28	

Timucin spins the spinner 200 times.

Work out an estimate for the number of times the spinner lands on pink.

(Total for Question 2 is 3 marks)

PAST PAPER SOLUTION

Q#: 2 Marks: 3

TOPIC: **Probability Toolkit**

Jun 2023 P2H - Jun 2023 | P2H

WORKED SOLUTION

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1. The probability of pink is

The probability of pink is

$$1 - 0.32 - 0.13 - 0.28 = 0.27$$

2. Expected number of pink sweets

Expected number of pink sweets:

$$200 \times 0.27 = 54$$

FINAL ANSWER

54.

PAST PAPER QUESTION Q#: 3 Marks: 4

TOPIC: **Angles in Polygons & Parallel Lines**

Jun 2023 P2H - Jun 2023 | P2H

3 *ABCD* is a trapezium.

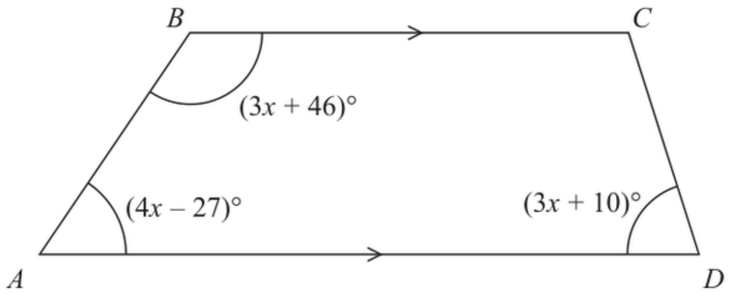

Diagram **NOT** accurately drawn

BC is parallel to *AD*

Find the size of the largest angle inside the trapezium.

.... °

(Total for Question 3 is 4 marks)

PAST PAPER SOLUTION

Q#: 3 Marks: 4

TOPIC: **Angles in Polygons & Parallel Lines**

Jun 2023 P2H - Jun 2023 | P2H

WORKED SOLUTION

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1. Find the gradient

Since $BC \parallel AD$, the co-interior angles on side AB add to $180°$:

$$(4x - 27) + (3x + 46) = 180$$

$$7x + 19 = 180$$

$$7x = 161$$

$$x = 23$$

2. Find the gradient

Now find the angles:

$$\angle A = 4(23) - 27 = 65°$$

$$\angle B = 3(23) + 46 = 115°$$

$$\angle D = 3(23) + 10 = 79°$$

3. Find the gradient

Also, co-interior angles on side CD add to $180°$:

$$\angle C = 180° - 79° = 101°$$

4. The largest angle is

The largest angle is $115°$.

FINAL ANSWER
$115°$.

PAST PAPER QUESTION

Q#: 4 Marks: 4

TOPIC: **Graphs of Functions**

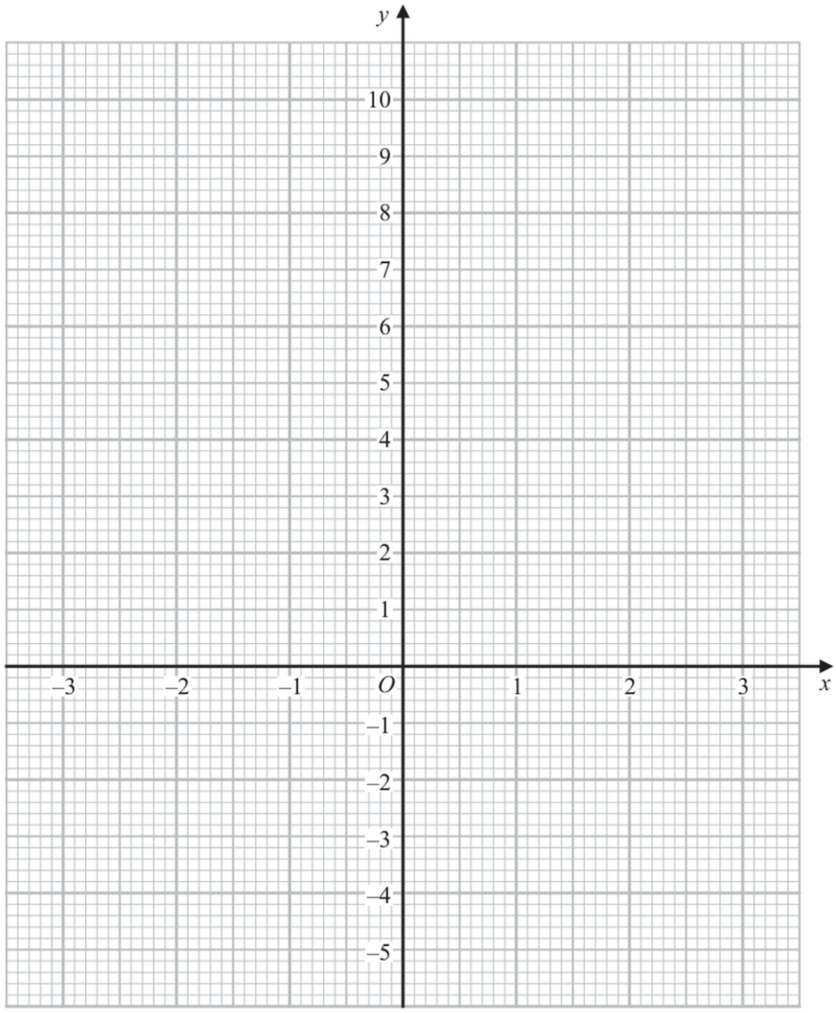
Jun 2023 P2H - Jun 2023 | P2H

4 (a) Complete the table of values for $y = x^2 - x - 4$

x	–3	–2	–1	0	1	2	3
y		2			–4		

(2)

(b) On the grid below, draw the graph of $y = x^2 - x - 4$ for values of x from –3 to 3



(2)

(Total for Question 4 is 4 marks)

PAST PAPER SOLUTION

Q#: 4 Marks: 4

TOPIC: **Graphs of Functions**

Jun 2023 P2H - Jun 2023 | P2H

WORKED SOLUTION

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1. Solve quadratic equation

$$y = x^2 - x - 4$$

2. The completed table is

The completed table is:

x	-3	-2	-1	0	1	2	3
y	8	2	-2	-4	-4	-2	2

3. Plot the points and join

Plot the points and join them with a smooth U-shaped curve.

FINAL ANSWER

Missing values are $8, -2, -4, -2, 2$.

PAST PAPER QUESTION

Q#: 5 Marks: 4

TOPIC: **Ratio Toolkit**

Jun 2023 P2H - Jun 2023 | P2H

5 Nancy has some coins with a total value of 85 pence.
She has only 2 pence coins and 5 pence coins.
The ratio

$$\text{number of 2 pence coins} : \text{number of 5 pence coins} = 1:3$$

Nancy has more 5 pence coins than 2 pence coins.

How many more?

(Total for Question 5 is 4 marks)

PAST PAPER SOLUTION

Q#: 5 Marks: 4

TOPIC: **Ratio Toolkit**

Jun 2023 P2H - Jun 2023 | P2H

WORKED SOLUTION

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1. Let the number of pence

Let the number of 2 pence coins be x.

2. Split the ratio

Then the number of 5 pence coins is $3x$.

3. Total value

Total value:

$$2x + 5(3x) = 85$$

$$17x = 85$$

$$x = 5$$

4. Split the ratio

So there are 5 two-pence coins and 15 five-pence coins.

$$15 - 5 = 10$$

FINAL ANSWER

10 more 5 pence coins

PAST PAPER QUESTION

Q#: 6 Marks: 2

TOPIC: **Powers, Roots & Standard Form**

Jun 2023 P2H - Jun 2023 | P2H

6 (a) Write 76 000 000 in standard form.

....
(1)

(b) Write 5.4×10^{-4} as an ordinary number.

....
(1)

(Total for Question 6 is 2 marks)

PAST PAPER SOLUTION

Q#: 6 Marks: 2

TOPIC: **Powers, Roots & Standard Form**

Jun 2023 P2H - Jun 2023 | P2H

WORKED SOLUTION

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1. Convert standard form

$$76000000 = 7.6 \times 10^7$$

$$5.4 \times 10^{-4} = 0.00054$$

FINAL ANSWER

(a) 7.6×10^7, (b) 0.00054

PAST PAPER QUESTION

Q#: 7 **Marks: 3**

TOPIC: **Circle Theorems**

Jun 2023 P2H - Jun 2023 | P2H

7

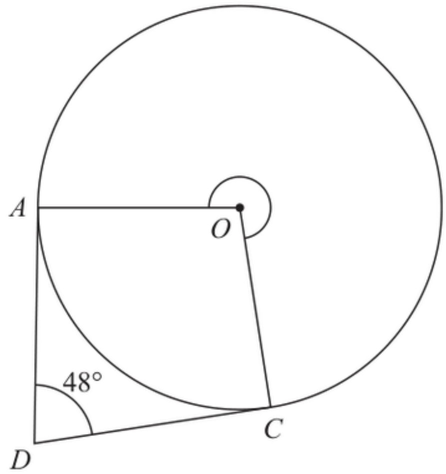

Diagram **NOT** accurately drawn

A and C are points on a circle, centre O
DA is the tangent to the circle at A and DC is the tangent to the circle at C

Angle $ADC = 48°$

Work out the size of reflex angle AOC

.... °

(Total for Question 7 is 3 marks)

PAST PAPER SOLUTION

Q#: 7 Marks: 3

TOPIC: **Circle Theorems**

Jun 2023 P2H - Jun 2023 | P2H

WORKED SOLUTION

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1. Use circle theorem

A radius is perpendicular to a tangent, so

$$\angle OAD = 90^\circ$$

2. Use trigonometry

and

$$\angle OCD = 90^\circ$$

3. Use trigonometry

In quadrilateral $AOCD$,

$$\angle AOC = 360^\circ - 90^\circ - 90^\circ - 48^\circ = 132^\circ$$

4. The question asks for the

The question asks for the reflex angle:

$$360^\circ - 132^\circ = 228^\circ$$

FINAL ANSWER

228°.

PAST PAPER QUESTION

Q#: 8 Marks: 3

TOPIC: **Compound Interest & Depreciation**

Jun 2023 P2H - Jun 2023 | P2H

8 Charlotte buys a painting for \$680
The value of the painting increases by 4% each year.

Work out the value of the painting at the end of 3 years.
Give your answer correct to the nearest \$

\$...

(Total for Question 8 is 3 marks)

PAST PAPER SOLUTION

Q#: 8 Marks: 3

TOPIC: **Compound Interest & Depreciation**

Jun 2023 P2H - Jun 2023 | P2H

WORKED SOLUTION

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1. A increase each year gives

A 4% increase each year gives a multiplier of

$$1.04$$

2. Use compound interest

After 3 years, the value is

$$680(1.04)^3$$

$$= 680(1.124864)$$

$$= 764.90752$$

3. Correct to the nearest dollar

Correct to the nearest dollar:

$$765$$

FINAL ANSWER

\$765

PAST PAPER QUESTION

Q#: 9 Marks: 3

TOPIC: **Standard & Compound Units**

Jun 2023 P2H - Jun 2023 | P2H

9 Change a speed of 27 kilometres per hour to a speed in metres per second.

.... m/s

(Total for Question 9 is 3 marks)

PAST PAPER SOLUTION

Q#: 9 Marks: 3

TOPIC: **Standard & Compound Units**

Jun 2023 P2H - Jun 2023 | P2H

WORKED SOLUTION

Dr. Eslam Ahmed

1. Convert km/h to m/s divide

To convert km/h to m/s, divide by 3.6.

$$27 \div 3.6 = 7.5$$

FINAL ANSWER

7.5 m/s.

PAST PAPER QUESTION

Q#: 10 Marks: 4

TOPIC: **Statistics Toolkit**

Jun 2023 P2H - Jun 2023 | P2H

10 Team **A** and Team **B** take part in a quiz league.

After 11 rounds, Team **A** has a mean score per round of 17
After 9 rounds, Team **B** has a mean score per round of 18

Both teams take part in a further round.
After this round, both teams have a mean score per round of 18.5

In the further round, Team **A** scored more points than Team **B**.

How many more?

(Total for Question 10 is 4 marks)

PAST PAPER SOLUTION

Q#: 10 Marks: 4

TOPIC: **Statistics Toolkit**

Jun 2023 P2H - Jun 2023 | P2H

WORKED SOLUTION

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1. Team A

Team A:

$$11 \times 17 = 187$$

2. Estimate the value

After the next round:

$$12 \times 18.5 = 222$$

$$\text{next score} = 222 - 187 = 35$$

3. Team B

Team B:

$$9 \times 18 = 162$$

4. Estimate the value

After the next round:

$$10 \times 18.5 = 185$$

$$\text{next score} = 185 - 162 = 23$$

5. Difference

Difference:

$$35 - 23 = 12$$

FINAL ANSWER

12.

PAST PAPER QUESTION

Q#: 11 Marks: 3

Topic: **Angles in Polygons & Parallel Lines**

Jun 2023 P2H - Jun 2023 | P2H

11 Here is a 9-sided regular polygon *ABCDEFGHJ*, with centre *O*

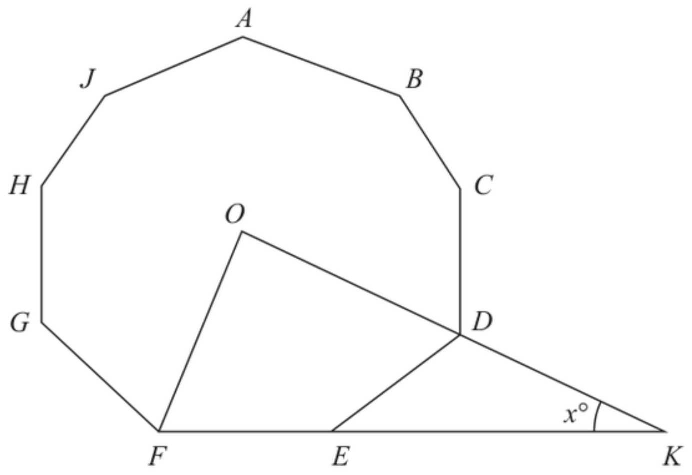

Diagram **NOT** accurately drawn

ODK and *FEK* are straight lines.

Work out the value of x

$x =$ ....

(Total for Question 11 is 3 marks)

PAST PAPER SOLUTION

Q#: 11 Marks: 3

TOPIC: **Angles in Polygons & Parallel Lines**

Jun 2023 P2H - Jun 2023 | P2H

WORKED SOLUTION

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1. Find the gradient

For a regular 9-sided polygon, the angle at the centre between adjacent vertices is

$$\frac{360}{9} = 40^\circ$$

2. Find the gradient

The side FE is perpendicular to the radius drawn to the midpoint of side FE.

3. From to this midpoint radius

From OD to this midpoint radius is

$$40^\circ + 20^\circ = 60^\circ$$

4. Find the gradient

So the angle between OD and the side FE is

$$90^\circ - 60^\circ = 30^\circ$$

5. Find the gradient

Since ODK and FEK are straight lines, this is the angle x.

FINAL ANSWER

$x = 30^\circ$.

PAST PAPER QUESTION

Q#: 12 Marks: 4

TOPIC: **Right-Angled Triangles - Pythagoras & Trigonometry**

Jun 2023 P2H - Jun 2023 | P2H

12 The diagram shows right-angled triangle *ABD*

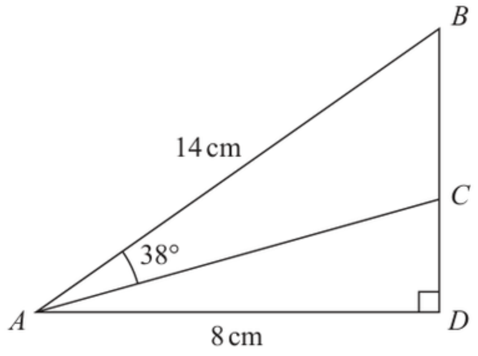

Diagram **NOT** accurately drawn

$AB = 14$ cm $\quad\quad AD = 8$ cm

C is the point on *BD* such that angle $BAC = 38°$

Work out the length of *CD*
Give your answer correct to 3 significant figures.

.... cm

(Total for Question 12 is 4 marks)

PAST PAPER SOLUTION

Q#: 12 Marks: 4

TOPIC: **Right-Angled Triangles - Pythagoras & Trigonometry**

Jun 2023 P2H - Jun 2023 | P2H

WORKED SOLUTION

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1. Use trigonometry

First find angle DAB:

$$\cos(\angle DAB) = \frac{8}{14}$$

$$\angle DAB = 55.150\ldots^\circ$$

2. Given

Given

$$\angle BAC = 38^\circ$$

3. Use trigonometry

So

$$\angle CAD = 55.150\ldots^\circ - 38^\circ = 17.150\ldots^\circ$$

4. Use trigonometry

In right-angled triangle ACD,

$$\tan(17.150\ldots^\circ) = \frac{CD}{8}$$

$$CD = 2.468\ldots$$

FINAL ANSWER

2.47 cm.

PAST PAPER QUESTION Q#: 13 Marks: 4

TOPIC: **Probability Diagrams - Venn & Tree Diagrams**

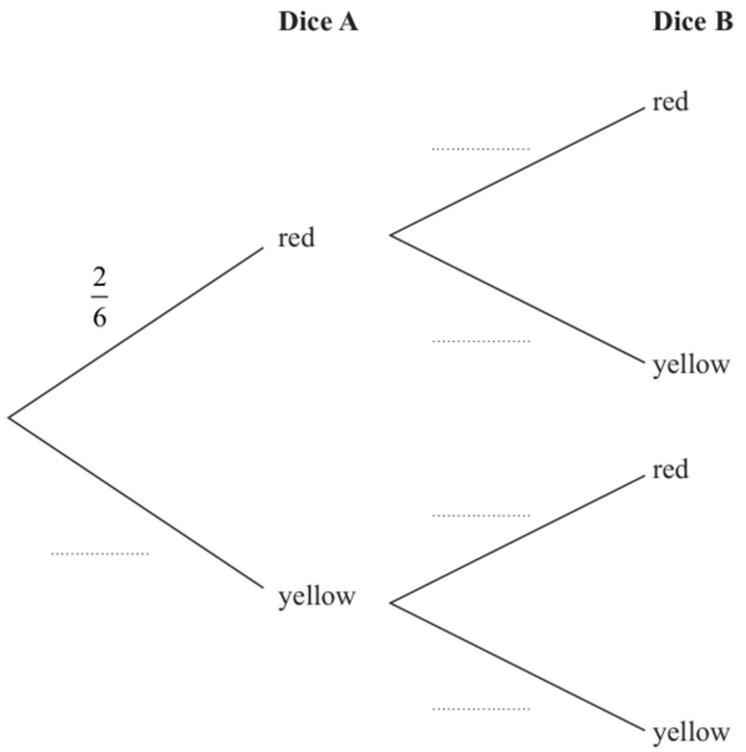
Jun 2023 P2H - Jun 2023 | P2H

13 Narin has two fair 6-sided dice.

Dice **A** has 2 red faces and 4 yellow faces.
Dice **B** has 1 red face and 5 yellow faces.

Narin is going to throw each dice once.

(a) Complete the probability tree diagram.



(2)

(b) Work out the probability that both dice land on yellow.

....

(2)

(Total for Question 13 is 4 marks)

PAST PAPER SOLUTION Q#: 13 Marks: 4

TOPIC: **Probability Diagrams - Venn & Tree Diagrams**

Jun 2023 P2H - Jun 2023 | P2H

WORKED SOLUTION

Dr. Eslam Ahmed

1. Dice A has yellow faces

Dice A has 4 yellow faces out of 6.
Dice B has 5 yellow faces out of 6.

$$P(\text{both yellow}) = \frac{4}{6} \times \frac{5}{6}$$

$$= \frac{5}{9}$$

FINAL ANSWER

$\frac{5}{9}$.

PAST PAPER QUESTION

Q#: 14 Marks: 4

TOPIC: **Circle Theorems**

Jun 2023 P2H - Jun 2023 | P2H

14

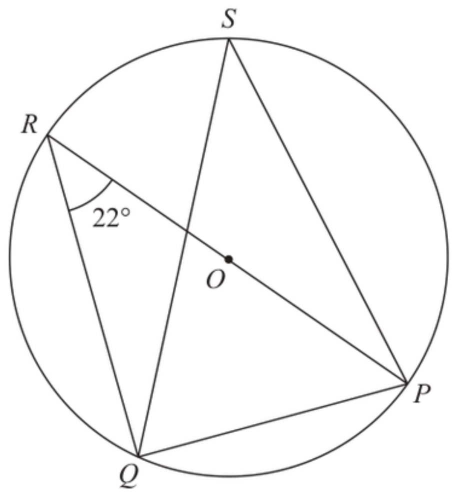


Diagram **NOT** accurately drawn

P, Q, R and S are points on a circle, centre O
ROP is a diameter of the circle.
Angle $PRQ = 22°$

(a) (i) Find the size of angle RQP

.... °

(1)

(ii) Give a reason for your answer.

....

(1)

(b) (i) Find the size of angle PSQ

.... °

(1)

(ii) Give a reason for your answer.

....

(1)

(Total for Question 14 is 4 marks)

PAST PAPER SOLUTION

Q#: 14 Marks: 4

TOPIC: **Circle Theorems**

Jun 2023 P2H - Jun 2023 | P2H

WORKED SOLUTION

Dr. Eslam Ahmed

1. Use trigonometry

(a) Since RP is a diameter,

$$\angle RQP = 90^\circ$$

2. Reason the angle in a

Reason: the angle in a semicircle is 90°.

3. (b) Angles and stand on

(b) Angles PRQ and PSQ stand on the same chord PQ, so they are equal.

$$\angle PSQ = 22^\circ$$

FINAL ANSWER

$\angle RQP = 90^\circ$, and $\angle PSQ = 22^\circ$.

PAST PAPER QUESTION Q#: 15 Marks: 7

TOPIC: **Algebraic Roots & Indices**

Jun 2023 P2H - Jun 2023 | P2H

15 (a) Simplify fully $(32a^{15})^{\frac{3}{5}}$

....
(2)

(b) Express $\left(\dfrac{1}{10x}\right)^{-3}$ in the form px^n where p and n are integers.

....
(2)

(c) Solve $\dfrac{1-2y}{3} = \dfrac{4}{5} - \dfrac{2y-1}{2}$

Show clear algebraic working.

$y =$ ....
(3)

(Total for Question 15 is 7 marks)

PAST PAPER SOLUTION

Q#: 15 Marks: 7

TOPIC: **Algebraic Roots & Indices**

Jun 2023 P2H - Jun 2023 | P2H

WORKED SOLUTION

Dr. Eslam Ahmed

1. Solve equation

For part (a),

$$(32a^{15})^{3/5} = 32^{3/5}a^{15\times 3/5}$$

$$= 8a^9$$

2. Solve equation

For part (b),

$$\left(\frac{1}{10x}\right)^{-3} = (10x)^3 = 1000x^3$$

3. Solve equation

For part (c),

$$\frac{1-2y}{3} = \frac{4}{5} - \frac{2y-1}{2}$$

4. Multiply by

Multiply by 30:

$$10(1-2y) = 24 - 15(2y-1)$$

$$10 - 20y = 39 - 30y$$

$$10y = 29$$

$$y = \frac{29}{10}$$

FINAL ANSWER

(a) $8a^9$, (b) $1000x^3$, (c) $y = \frac{29}{10}$

PAST PAPER QUESTION

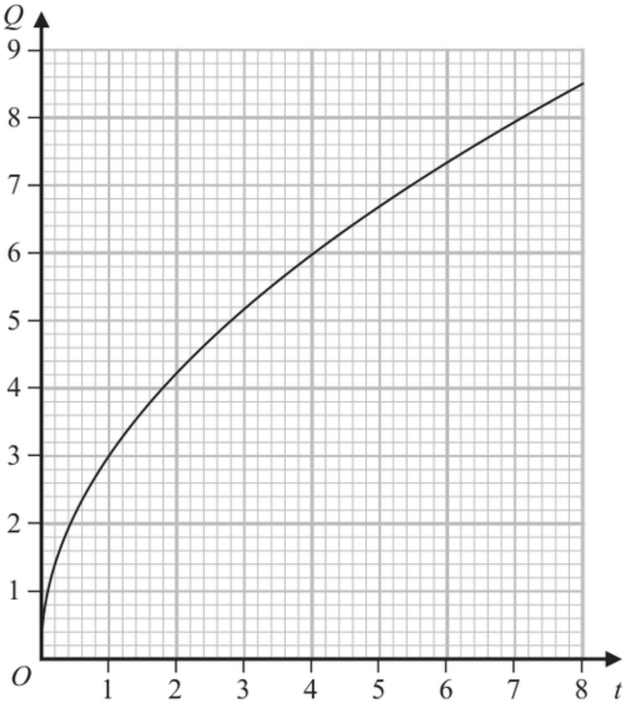
Q#: 16 Marks: 5

TOPIC: **Direct & Inverse Proportion**

Jun 2023 P2H - Jun 2023 | P2H

16 Q is directly proportional to $\sqrt{t}$

The graph shows the relationship between Q and t for $0 < t < 8$



(a) Find a formula for Q in terms of t

....

(3)

Q is increased by 20%

(b) Find the percentage increase in t

.... %

(2)

(Total for Question 16 is 5 marks)

PAST PAPER SOLUTION Q#: 16 Marks: 5

TOPIC: **Direct & Inverse Proportion**

Jun 2023 P2H - Jun 2023 | P2H

WORKED SOLUTION

Dr. Eslam Ahmed

1. Find inverse function

$$Q \propto \sqrt{t}$$

2. Find inverse function

So

$$Q = k\sqrt{t}$$

3. Read the graph

From the graph, when $t = 4$, $Q = 6$.

$$6 = k\sqrt{4}$$

$$6 = 2k$$

$$k = 3$$

4. Rearrange formula

So the formula is

$$Q = 3\sqrt{t}$$

5. Find inverse function

For part (b), Q is increased by 20%, so Q is multiplied by 1.2.

6. Find inverse function

Since $Q \propto \sqrt{t}$,

$$t \propto Q^2$$

7. Find inverse function

So t is multiplied by

$$1.2^2 = 1.44$$

8. This is a increase

This is a 44% increase.

FINAL ANSWER

(a) $Q = 3\sqrt{t}$, (b) 44%

PAST PAPER QUESTION Q#: 17 Marks: 7

TOPIC: **Rearranging Formulas**

Jun 2023 P2H - Jun 2023 | P2H

17 (a) Expand and simplify $(x + 6)(3x - 2)(x + 6)$

....

(3)

(b) Make e the subject of $w = \sqrt{\dfrac{e + g}{ef - d}}$

....

(4)

(Total for Question 17 is 7 marks)

PAST PAPER SOLUTION Q#: 17 Marks: 7

TOPIC: **Rearranging Formulas**

Jun 2023 P2H - Jun 2023 | P2H

WORKED SOLUTION *Dr. Eslam Ahmed*

1. (a)

(a)

$$(x+6)(3x-2)(x+6) = (x+6)^2(3x-2)$$

$$= (x^2+12x+36)(3x-2)$$

$$= 3x^3+34x^2+84x-72$$

2. (b)

(b)

$$w = \sqrt{\frac{e+g}{ef-d}}$$

$$w^2 = \frac{e+g}{ef-d}$$

$$w^2(ef-d) = e+g$$

$$e(fw^2-1) = g+dw^2$$

FINAL ANSWER

(a) $3x^3+34x^2+84x-72$, (b) $e = \frac{g+dw^2}{fw^2-1}$

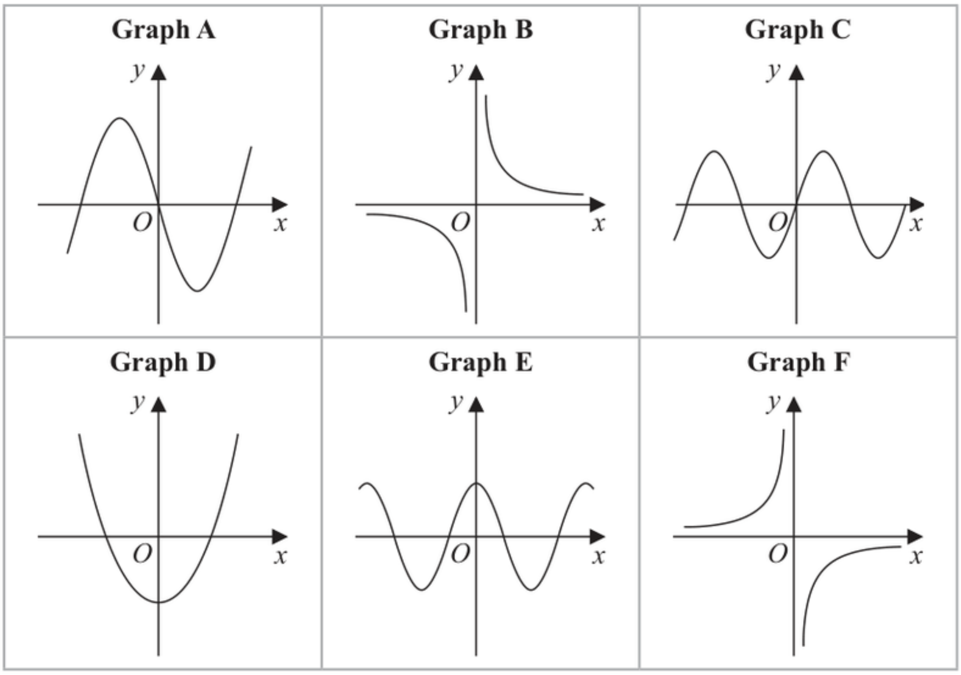
PAST PAPER QUESTION

Q#: 18 Marks: 3

TOPIC: **Graphs of Functions**

Jun 2023 P2H - Jun 2023 | P2H

18 Here are 6 graphs.



Complete the table below with the letter of the graph that could represent each given equation.

Write your answers on the dotted lines.

Equation	**Graph**
$y = \sin x$	....
$y = -\dfrac{3}{x}$	....
$y = 4x^3 - 5x$	....

(Total for Question 18 is 3 marks)

PAST PAPER SOLUTION

Q#: 18 Marks: 3

TOPIC: **Graphs of Functions**

Jun 2023 P2H - Jun 2023 | P2H

WORKED SOLUTION

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1. is the periodic sine shaped

$$y = \sin x$$

is the periodic sine-shaped graph, so it is **Graph C**.

2. Read the graph

$$y = \frac{3}{x}$$

is the positive reciprocal graph, so it is **Graph B**.

3. Find turning point

$$y = 4x^3 - 5x$$

is a cubic graph with two turning points, so it is **Graph A**.

FINAL ANSWER

$C,\ B,\ A.$

PAST PAPER QUESTION

Q#: 19 Marks: 3

TOPIC: **Completing the Square**

Jun 2023 P2H - Jun 2023 | P2H

19 Express $3x^2 - 6x + 5$ in the form $a(x - b)^2 + c$

(Total for Question 19 is 3 marks)

PAST PAPER SOLUTION

Q#: 19 Marks: 3

TOPIC: **Completing the Square**

Jun 2023 P2H - Jun 2023 | P2H

WORKED SOLUTION

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1. Calculate value

$$3x^2 - 6x + 5 = 3(x^2 - 2x) + 5$$

$$= 3((x-1)^2 - 1) + 5$$

$$= 3(x-1)^2 + 2$$

FINAL ANSWER

$3(x-1)^2 + 2$

PAST PAPER QUESTION

Q#: 20 Marks: 3

TOPIC: **Combined & Conditional Probability**

Jun 2023 P2H - Jun 2023 | P2H

20 There are 12 counters in a bag.

3 of the counters are red
9 of the counters are green

Ameya, Jack and Ella each take at random one counter from the bag.

Work out the probability that at least one red counter is still in the bag.

(Total for Question 20 is 3 marks)

PAST PAPER SOLUTION

Q#: 20 Marks: 3

TOPIC: **Combined & Conditional Probability**

Jun 2023 P2H - Jun 2023 | P2H

WORKED SOLUTION

Dr. Eslam Ahmed

1. There are 3 red counters

There are 3 red counters.

2. At least one red counter

At least one red counter is still in the bag unless all 3 red counters are taken.

3. Use the complement

So use the complement:

$$P(\text{all 3 red are taken}) = \frac{3}{12} \cdot \frac{2}{11} \cdot \frac{1}{10}$$

$$= \frac{6}{1320}$$

$$= \frac{1}{220}$$

4. Evaluate fraction

Therefore,

$$P(\text{at least one red is still in the bag}) = 1 - \frac{1}{220}$$

$$= \frac{219}{220}$$

FINAL ANSWER

$\frac{219}{220}$

PAST PAPER QUESTION

Q#: 20 Marks: 3

TOPIC: **Combined & Conditional Probability**

Jun 2023 P2H - Jun 2023 | P2H

20 There are 12 counters in a bag.

3 of the counters are red
9 of the counters are green

Ameya, Jack and Ella each take at random one counter from the bag.

Work out the probability that at least one red counter is still in the bag.

(Total for Question 20 is 3 marks)

PAST PAPER SOLUTION Q#: 20 Marks: 3

TOPIC: **Combined & Conditional Probability**

Jun 2023 P2H - Jun 2023 | P2H

WORKED SOLUTION

Dr. Eslam Ahmed

1. There are 3 red counters

There are 3 red counters.

2. At least one red counter

At least one red counter is still in the bag unless all 3 red counters are taken.

3. Use the complement

So use the complement:

$$P(\text{all 3 red are taken}) = \frac{3}{12} \cdot \frac{2}{11} \cdot \frac{1}{10}$$

$$= \frac{6}{1320}$$

$$= \frac{1}{220}$$

4. Evaluate fraction

Therefore,

$$P(\text{at least one red is still in the bag}) = 1 - \frac{1}{220}$$

$$= \frac{219}{220}$$

FINAL ANSWER

$\frac{219}{220}$

PAST PAPER QUESTION

Q#: 21 Marks: 5

TOPIC: **Simultaneous Equations**

Jun 2023 P2H - Jun 2023 | P2H

21 Solve the simultaneous equations

$$2x^2 + 3y^2 = 11$$
$$x = 3y - 1$$

Show clear algebraic working.

(Total for Question 21 is 5 marks)

PAST PAPER SOLUTION

Q#: 21 **Marks: 5**

TOPIC: **Simultaneous Equations**

Jun 2023 P2H - Jun 2023 | P2H

WORKED SOLUTION

Dr. Eslam Ahmed

1. Calculate value

$$2x^2 + 3y^2 = 11$$

$$x = 3y - 1$$

2. Substitute

Substitute:

$$2(3y-1)^2 + 3y^2 = 11$$

$$18y^2 - 12y + 2 + 3y^2 = 11$$

$$21y^2 - 12y - 9 = 0$$

$$7y^2 - 4y - 3 = 0$$

$$(7y+3)(y-1) = 0$$

3. Evaluate fraction

So

$$y = -\frac{3}{7} \quad \text{or} \quad y = 1$$

4. Find

Find x:

$$y = -\frac{3}{7} \Rightarrow x = -\frac{16}{7}$$

$$y = 1 \Rightarrow x = 2$$

FINAL ANSWER

$(x, y) = \left(-\frac{16}{7}, -\frac{3}{7}\right)$ or $(2, 1)$

PAST PAPER QUESTION

Topic: **Simultaneous Equations**

Jun 2023 P2H - Jun 2023 | P2H

Q#: 21 Marks: 5

21 Solve the simultaneous equations

$$2x^2 + 3y^2 = 11$$
$$x = 3y - 1$$

Show clear algebraic working.

(Total for Question 21 is 5 marks)

PAST PAPER SOLUTION

Q#: 21 Marks: 5

TOPIC: **Simultaneous Equations**

Jun 2023 P2H - Jun 2023 | P2H

WORKED SOLUTION

Dr. Eslam Ahmed

1. Calculate value

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Substitute:

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FINAL ANSWER

$(x, y) = \left(-\frac{16}{7}, -\frac{3}{7}\right)$ or $(2, 1)$

PAST PAPER QUESTION

Q#: 22 Marks: 5

TOPIC: Sine, Cosine Rule & Area of Triangles

Jun 2023 P2H - Jun 2023 | P2H

22 The diagram shows a triangle *ABC* and a flagpole *BF*

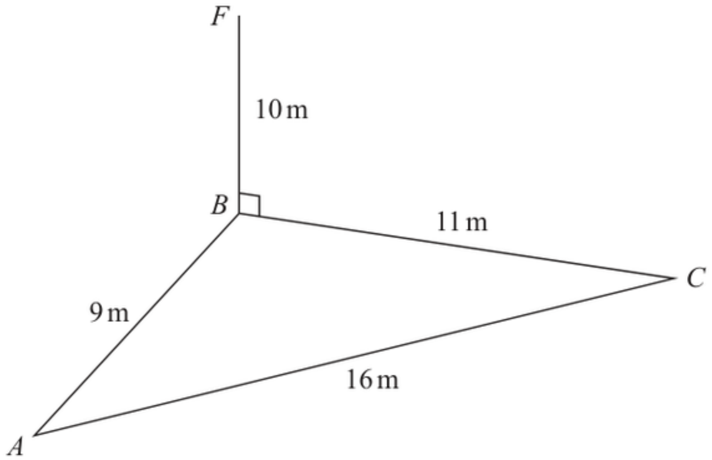


Diagram **NOT** accurately drawn

A, *B* and *C* are points on horizontal ground.

BF is vertical.

$AB = 9\,\text{m}$ $BC = 11\,\text{m}$ $AC = 16\,\text{m}$ $BF = 10\,\text{m}$

D is the point on *AC* such that angle $BDC = 90°$

Work out the size of the angle of elevation of the point *F* from the point *D*
Give your answer correct to one decimal place.

.... °

(Total for Question 22 is 5 marks)

PAST PAPER SOLUTION Q#: 22 Marks: 5

TOPIC: **Sine, Cosine Rule & Area of Triangles**

Jun 2023 P2H - Jun 2023 | P2H

WORKED SOLUTION

Dr. Eslam Ahmed

1. Use trigonometry

In triangle ABC,

$$AB = 9, \qquad BC = 11, \qquad AC = 16$$

2. Use Heron s formula

Use Heron's formula.

$$s = \frac{9 + 11 + 16}{2} = 18$$

$$\text{area} = \sqrt{18(18-9)(18-11)(18-16)}$$

$$= \sqrt{18 \cdot 9 \cdot 7 \cdot 2} = 47.6235\ldots$$

3. is the foot of the

D is the foot of the perpendicular from B to AC, so BD is the height of triangle ABC.

$$\text{area} = \frac{1}{2} \times AC \times BD$$

$$47.6235\ldots = \frac{1}{2} \times 16 \times BD$$

$$BD = 5.9529\ldots$$

4. Use trigonometry

In right-angled triangle BDF,

$$\tan\theta = \frac{BF}{BD}$$

$$\tan\theta = \frac{10}{5.9529\ldots}$$

$$\theta = \tan^{-1}\left(\frac{10}{5.9529\ldots}\right) = 59.2349\ldots^\circ$$

FINAL ANSWER

$\boxed{59.2^\circ}$

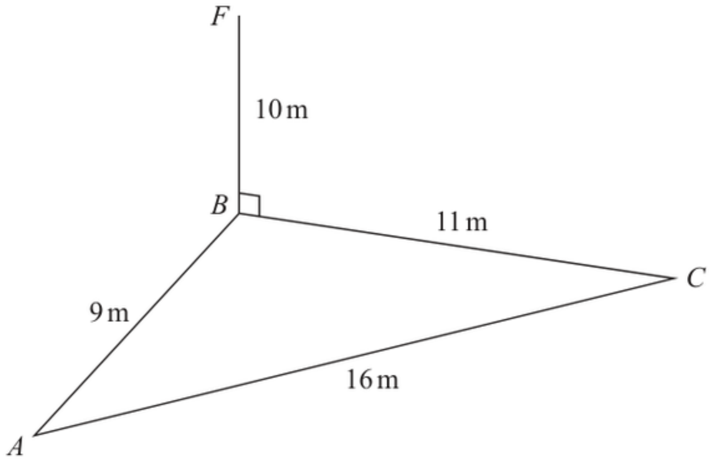
PAST PAPER QUESTION

Q#: 22 Marks: 5

TOPIC: **Sine, Cosine Rule & Area of Triangles**

Jun 2023 P2H - Jun 2023 | P2H

22 The diagram shows a triangle ABC and a flagpole BF



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D is the point on AC such that angle $BDC = 90°$

Work out the size of the angle of elevation of the point F from the point D
Give your answer correct to one decimal place.

.... °

(Total for Question 22 is 5 marks)

PAST PAPER SOLUTION

Q#: 22 Marks: 5

TOPIC: **Sine, Cosine Rule & Area of Triangles**

Jun 2023 P2H - Jun 2023 | P2H

WORKED SOLUTION

Dr. Eslam Ahmed

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FINAL ANSWER

$\boxed{59.2^\circ}$

PAST PAPER QUESTION

Q#: 23 Marks: 4

TOPIC: **Surds**

Jun 2023 P2H - Jun 2023 | P2H

23 The diagram shows a cuboid with a square cross section.

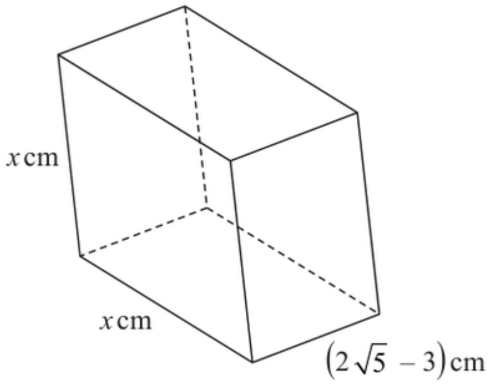


Diagram **NOT** accurately drawn

The volume of the cuboid is $(13 + 6\sqrt{5})\text{ cm}^3$

Without using a calculator, find the value of x
Give your answer in the form $a + \sqrt{b}$ where a and b are integers.
Show your working clearly.

$x =$ ....

(Total for Question 23 is 4 marks)

PAST PAPER SOLUTION Q#: 23 Marks: 4

TOPIC: **Surds**

Jun 2023 P2H - Jun 2023 | P2H

WORKED SOLUTION *Dr. Eslam Ahmed*

1. The volume is

The volume is

$$x^2(2\sqrt{5}-3) = 13 + 6\sqrt{5}$$

2. Simplify surd

So

$$x^2 = \frac{13+6\sqrt{5}}{2\sqrt{5}-3}$$

3. Rationalise

Rationalise:

$$x^2 = \frac{(13+6\sqrt{5})(2\sqrt{5}+3)}{(2\sqrt{5})^2-3^2}$$

$$= \frac{26\sqrt{5}+39+60+18\sqrt{5}}{20-9}$$

$$= \frac{99+44\sqrt{5}}{11} = 9+4\sqrt{5}$$

4. Simplify surd

Since

$$(\sqrt{5}+2)^2 = 5+4+4\sqrt{5} = 9+4\sqrt{5}$$

$$x = \sqrt{5}+2$$

FINAL ANSWER

$2+\sqrt{5}$

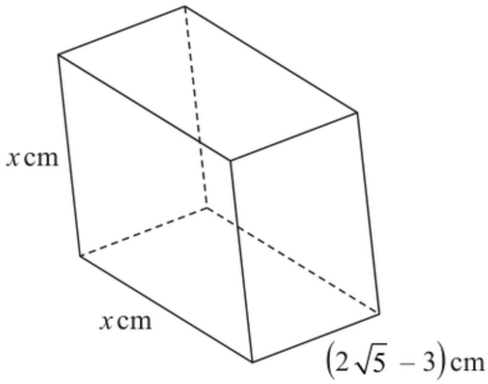
PAST PAPER QUESTION

Q#: 23 Marks: 4

Topic: **Surds**

Jun 2023 P2H - Jun 2023 | P2H

23 The diagram shows a cuboid with a square cross section.



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The volume of the cuboid is $(13 + 6\sqrt{5})\,\text{cm}^3$

Without using a calculator, find the value of x
Give your answer in the form $a + \sqrt{b}$ where a and b are integers.
Show your working clearly.

$x =$ ....

(Total for Question 23 is 4 marks)

PAST PAPER SOLUTION Q#: 23 Marks: 4

TOPIC: **Surds**

Jun 2023 P2H - Jun 2023 | P2H

WORKED SOLUTION

Dr. Eslam Ahmed

1. The cuboid has square cross

The cuboid has square cross section, so its volume is

$$x^2(2\sqrt{5}-3)$$

2. Given

Given

$$x^2(2\sqrt{5}-3) = 13+6\sqrt{5}$$

3. Use trigonometry

So

$$x^2 = \frac{13+6\sqrt{5}}{2\sqrt{5}-3}$$

4. Rationalise the denominator

Rationalise the denominator:

$$x^2 = \frac{13+6\sqrt{5}}{2\sqrt{5}-3}\cdot\frac{2\sqrt{5}+3}{2\sqrt{5}+3}$$
$$x^2 = \frac{(13+6\sqrt{5})(2\sqrt{5}+3)}{20-9}$$
$$x^2 = \frac{99+44\sqrt{5}}{11}$$
$$x^2 = 9+4\sqrt{5}$$

5. Use trigonometry

Now

$$(2+\sqrt{5})^2 = 4+4\sqrt{5}+5 = 9+4\sqrt{5}$$

6. Use trigonometry

Therefore

$$x = 2+\sqrt{5}$$

FINAL ANSWER

$\boxed{2+\sqrt{5}}$

PAST PAPER QUESTION Q#: 24 Marks: 6

TOPIC: **Coordinate Geometry**

Jun 2023 P2H - Jun 2023 | P2H

24 *ABCD* is a kite with $AB = AD$ and $CB = CD$

A is the point with coordinates $(-2, 10)$

B is the point with coordinates $\left(-\frac{27}{5}, 4\right)$

C is the point with coordinates $(4, -5)$

Work out the coordinates of *D*

(.......................... , ...)

(Total for Question 24 is 6 marks)

PAST PAPER SOLUTION

Q#: 24 Marks: 6

TOPIC: **Coordinate Geometry**

Jun 2023 P2H - Jun 2023 | P2H

WORKED SOLUTION

Dr. Eslam Ahmed

1. Calculate value

In the kite, $AB = AD$ and $CB = CD$, so AC is the line of symmetry.

2. Apply transformation

Therefore D is the reflection of B in the line AC.

$$A = (-2, 10), \qquad B = \left(-\frac{27}{5}, 4\right), \qquad C = (4, -5)$$

3. A direction vector for is

A direction vector for AC is

$$\begin{pmatrix} 4-(-2) \\ -5-10 \end{pmatrix} = \begin{pmatrix} 6 \\ -15 \end{pmatrix}$$

4. Use the simpler direction vector

Use the simpler direction vector

$$\begin{pmatrix} 2 \\ -5 \end{pmatrix}$$

5. Find the gradient

Let H be the foot of the perpendicular from B to AC.

$$\vec{AB} = \begin{pmatrix} -\frac{17}{5} \\ -6 \end{pmatrix}$$

6. The fraction along to is

The fraction along AC to H is

$$\frac{\vec{AB} \cdot \begin{pmatrix} 2 \\ -5 \end{pmatrix}}{2^2 + (-5)^2} = \frac{-\frac{34}{5} + 30}{29}$$
$$= \frac{\frac{116}{5}}{29} = \frac{4}{5}$$

7. Use matrices

So

$$H = A + \frac{4}{5}\begin{pmatrix} 2 \\ -5 \end{pmatrix}$$

$$H = \left(-2 + \frac{8}{5}, 10 - 4\right)$$

$$H = \left(-\frac{2}{5}, 6\right)$$

8. Find the midpoint

Since H is the midpoint of BD,

$$D = 2H - B$$

$$D = \left(-\frac{4}{5}, 12\right) - \left(-\frac{27}{5}, 4\right)$$

$$D = \left(\frac{23}{5}, 8\right)$$

FINAL ANSWER

$D = \left(\frac{23}{5}, 8\right)$

PAST PAPER QUESTION Q#: 24 Marks: 6

TOPIC: **Coordinate Geometry**

Jun 2023 P2H - Jun 2023 | P2H

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Work out the coordinates of *D*

(.......................... , ...)

(Total for Question 24 is 6 marks)

PAST PAPER SOLUTION Q#: 24 Marks: 6

TOPIC: **Coordinate Geometry**

Jun 2023 P2H - Jun 2023 | P2H

WORKED SOLUTION

Dr. Eslam Ahmed

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FINAL ANSWER

$D = \left(\frac{23}{5}, 8\right)$

PAST PAPER QUESTION

Q#: 25 Marks: 4

TOPIC: **Rounding, Estimation & Bounds**

Jun 2023 P2H - Jun 2023 | P2H

25 A solid sphere has a radius of 2.8 centimetres, correct to 1 decimal place.
The sphere has a mass of $M\pi$ grams, where $M = 260$ correct to 2 significant figures.

Work out the upper bound for the density of the sphere.
Give your answer in g/cm³ correct to 2 decimal places.
Show your working clearly.

.... g/cm^3

(Total for Question 25 is 4 marks)

PAST PAPER SOLUTION Q#: 25 Marks: 4

TOPIC: **Rounding, Estimation & Bounds**

Jun 2023 P2H - Jun 2023 | P2H

WORKED SOLUTION

Dr. Eslam Ahmed

1. Find upper bound

For the upper bound of density, use the upper bound of the mass and the lower bound of the volume.

2. The radius is cm correct

The radius is 2.8 cm correct to 1 decimal place, so the lower bound is
$r = 2.75$

3. Find upper bound

$M = 260$ correct to 2 significant figures, so the upper bound is
$M = 265$

4. The mass is and the

The mass is $M\pi$ and the volume of the sphere is $\frac{4}{3}\pi r^3$.

5. Rearrange formula

So
density $= \frac{M\pi}{\frac{4}{3}\pi r^3} = \frac{3M}{4r^3}$

6. Upper bound

Upper bound:
$\frac{3(265)}{4(2.75)^3} = 9.5567...$

FINAL ANSWER

9.56 g/cm^3 to 2 decimal places.

PAST PAPER QUESTION

Q#: 25 Marks: 4

TOPIC: **Rounding, Estimation & Bounds**

Jun 2023 P2H - Jun 2023 | P2H

25 A solid sphere has a radius of 2.8 centimetres, correct to 1 decimal place.
The sphere has a mass of $M\pi$ grams, where $M = 260$ correct to 2 significant figures.

Work out the upper bound for the density of the sphere.
Give your answer in g/cm³ correct to 2 decimal places.
Show your working clearly.

.... g/cm³

(Total for Question 25 is 4 marks)

PAST PAPER SOLUTION Q#: 25 Marks: 4

TOPIC: **Rounding, Estimation & Bounds**

Jun 2023 P2H - Jun 2023 | P2H

WORKED SOLUTION

Dr. Eslam Ahmed

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For the upper bound of density, use the upper bound of the mass and the lower bound of the volume.

2. The radius is cm correct

The radius is 2.8 cm correct to 1 decimal place, so the lower bound is
$r = 2.75$

3. Find upper bound

$M = 260$ correct to 2 significant figures, so the upper bound is
$M = 265$

4. The mass is given as

The mass is given as $M\pi$ grams, where $M = 265$ for the upper bound.

5. The volume of the sphere

The volume of the sphere is

$$\frac{4}{3}\pi r^3$$

6. Rearrange formula

So

$$\text{density} = \frac{M\pi}{\frac{4}{3}\pi r^3}$$

7. The cancels

The π cancels:

$$\text{density} = \frac{3M}{4r^3}$$

8. Upper bound

Upper bound:
$\dfrac{3(265)}{4(2.75)^3} = 9.5567...$

FINAL ANSWER

9.56 g/cm^3 to 2 decimal places.